# LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

## Member

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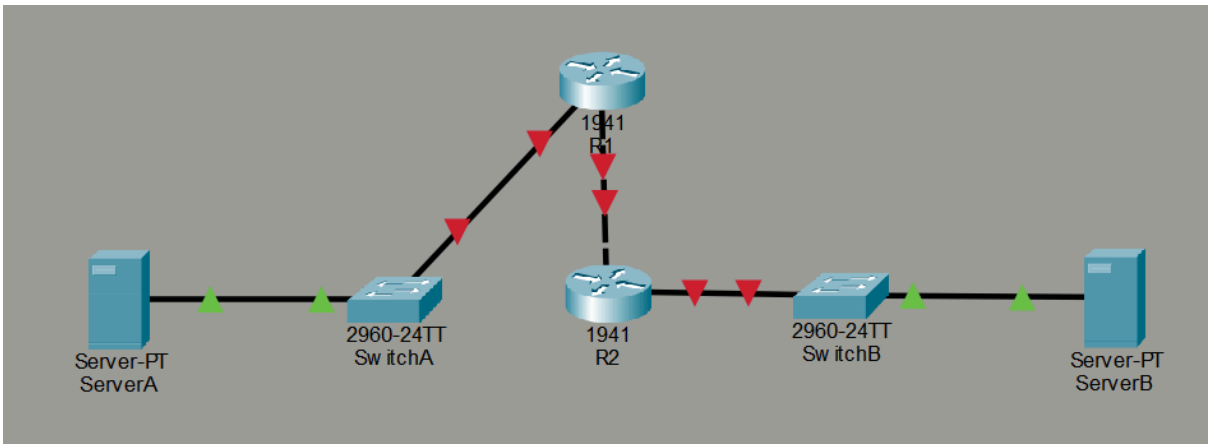
## TOPOLOGY

None

```
                    INTERNET
                       |
                   (DHCP)
                    R1 G0/0
                       |
                [NAT OUTSIDE]
                       |
               R1 S0/0/0
            100.10.10.1/30
                       |
               R2 S0/0/0
            100.10.10.2/30
                       |
           -----------------------
           |                     |
R1 G0/1 (LAN A)       R2 G0/1 (LAN B)
```

```
192.168.10.1            192.168.20.1
        |                       |
    ServerA                 ServerB
Week03 Microservices    Week03 Microservices
```

**ผลลัพธ์ :**

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# ADDRESS TABLE

| Device | Interface | IP | Role |
|---|---|---|---|
| R1 | G0/0 | DHCP | Internet |
| R1 | S0/0/0 | 100.10.10.1/30 | ISP |
| R1 | G0/1 | 192.168.10.1/24 | LAN A |
| R2 | S0/0/0 | 100.10.10.2/30 | ISP |
| R2 | G0/1 | 192.168.20.1/24 | LAN B |
| ServerA | eth0 | 192.168.10.10 | Microservices A |
| ServerB | eth0 | 192.168.20.10 | Microservices B |

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# PART 1 CISCO CONFIGURATION (STRICT)

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## R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/0
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

interface s0/0/0
 ip address 100.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

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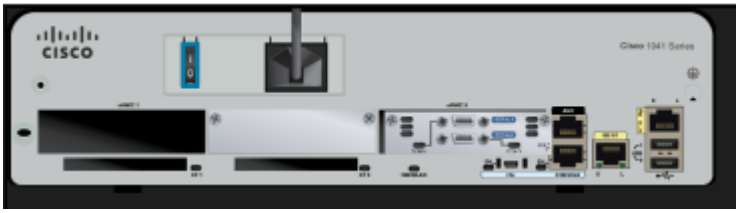
## ผลลัพธ์ของการ Config R1 :

**step 1 :** เพิ่ม port s0/0/0 ให้กับ R1 เพื่อให้ตรงกับ Config Code ของ Lab โดยใช้ HWIC-2T ใส่ไว้ที่ Port 0 ของ R1 โดยต้องปิด switch R1 ก่อนเพิ่มด้วย

**ก่อนเพิ่ม**

**หลังเพิ่ม**




**Step 2 :** จากนั้นทำการ Config R1 ตาม Code Lab

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#ip address dhcp
R1(config-if)#ip nat outside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R1(config-if)#interface g0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#interface s0/0/0
R1(config-if)#ip address 100.10.10.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
```

## R2 — Branch Router

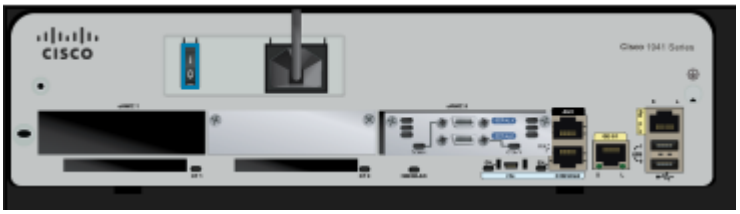
```
None
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/0/0
 ip address 100.10.10.2 255.255.255.252
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1

end
write memory
```

**ผลลัพธ์ของการ Config R2 :**

**step 1 :** เช่นเดียวกับ R1 ต้องเพิ่ม port s0/0/0 ให้กับ R2 เพื่อให้ตรงกับ Config Code ของ Lab โดยใช้ HWIC-2T ใส่ไว้ที่ Port 0 ของ R2 โดยต้องปิด switch R2 ก่อนเพิ่มด้วยเช่นกัน

**ก่อนเพิ่ม**

**หลังเพิ่ม**

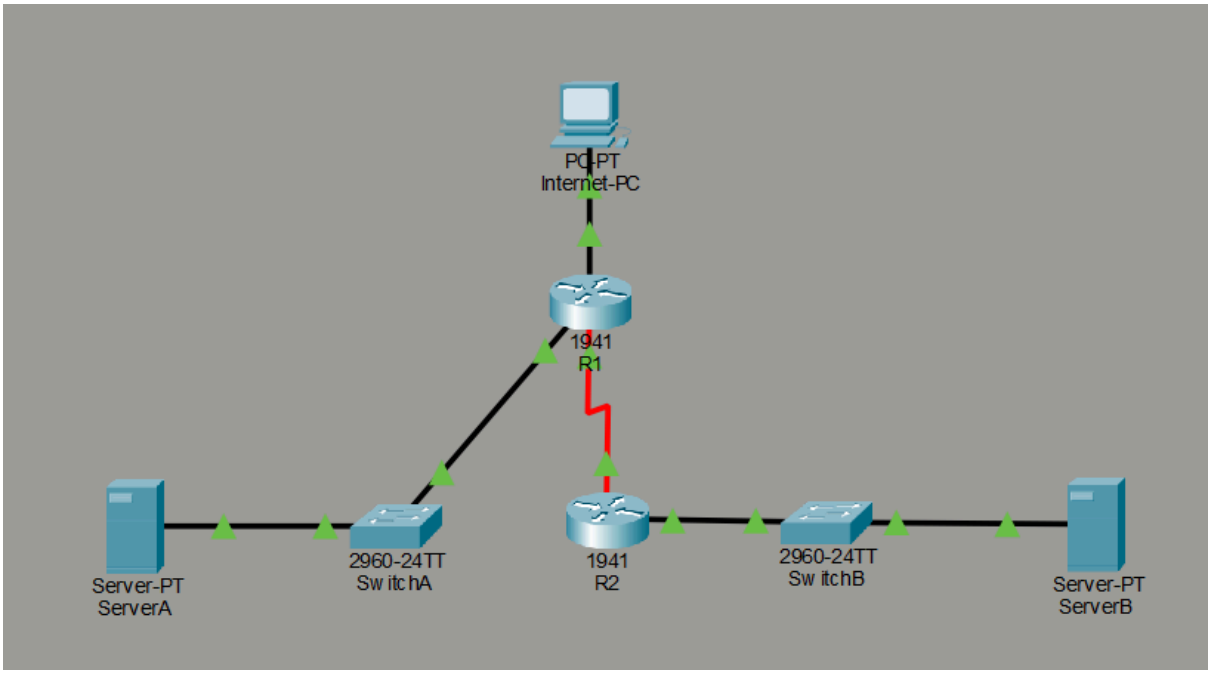



**Step 2 :** จากนั้นทำการ Config R2 ตาม Code Lab

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#interface s0/0/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R2(config-if)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
```

---

## ภาพรวม :

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# PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

---

## 🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

ผลลัพธ์ :

```
🎯 Starting Phase 1 Microservices...
==================================================
🚀 Starting Upload Service on port 8001...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8002...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8003...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 8090...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!

🔍 Checking system health...
System status: healthy
   ✅ gateway: healthy
   ✅ upload: healthy
   ✅ processing: healthy
   ✅ ai: healthy

🧪 Running quick integration test...
📄 Created test file: test_data\sample.jpg
❌ Quick test failed: 422
   Response: {"detail":[{"type":"model_attributes_type","loc":["body","processing_options"],"msg":
rue, \"processing_operation\": \"thumbnail\", \"enable_ai_analysis\": true, \"ai_analysis_type\":

🎯 Phase 1 Microservices are ready!

Available endpoints:
  Gateway:    http://localhost:8090
  Upload:     http://localhost:8001
  Processing: http://localhost:8002
  AI:         http://localhost:8003

Test commands:
  curl -X POST -F 'file=@test_data/sample.jpg' http://localhost:8090/process-file
  pytest tests/test_phase1_compliance.py -v

🔄 Press Ctrl+C to stop all services...

📊 Service Status:
----------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
Gateway Service      🟢 Running
```

Services:

- Upload → 8000
- Processing → 8001
- AI → 8002
- Gateway → 9000

ServerA IP:

```
None
192.168.10.10
```

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## 🛠️ On ServerB (LAN B)

Same deployment:

```
Shell
python start_services.py
```

เนื่องจากเราใช้การจำลองใน simulate ที่เครื่องเดียวกัน ทำให้ไม่สามารถทำขั้นตอนนี้ได้

ServerB IP:

```
None
192.168.20.10
```

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# 🧪 PART 3 — WAN + INTERNET TESTING

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## 🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
None
curl http://192.168.20.10:8000/health
```

Should traverse:

ClientA → R1 → Serial ISP → R2 → ServerB

ผลลัพธ์ :

```
PS E:\Levto\Com Sci KKu\ปี 2 เทอม 2\Computer Netwrok\automation\week03-microservices> curl http://localhost:8000/health


StatusCode        : 200
StatusDescription : OK
Content           : {"status":"healthy","service":"upload-service","timestamp":"2026-03-29T18:09:30.159677"}
RawContent        : HTTP/1.1 200 OK
                    Content-Length: 88
                    Content-Type: application/json
                    Date: Sun, 29 Mar 2026 11:09:29 GMT
                    Server: uvicorn

                    {"status":"healthy","service":"upload-service","timestamp":"2026-03-29T18:0...
Forms             : {}
Headers           : {[Content-Length, 88], [Content-Type, application/json], [Date, Sun, 29 Mar 2026 11:09:29 GMT], [Server, uvicorn]}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 88
```

```
PS E:\Levto\Com Sci KKu\ปี 2 เทอม 2\Computer Netwrok\automation\week03-microservices> curl http://localhost:9000/health


StatusCode        : 200
StatusDescription : OK
Content           : {"status":"healthy","timestamp":"2026-03-29T18:11:57.069880","check_duration":1.224777,"services":{"gateway":{"url":"http://localhost:8090/health","status":"healthy"},"upload":{"url
                    ":"http://localhost...
RawContent        : HTTP/1.1 200 OK
                    Content-Length: 369
                    Content-Type: application/json
                    Date: Sun, 29 Mar 2026 11:11:55 GMT
                    Server: uvicorn

                    {"status":"healthy","timestamp":"2026-03-29T18:11:57.069880","check_durati...
Forms             : {}
Headers           : {[Content-Length, 369], [Content-Type, application/json], [Date, Sun, 29 Mar 2026 11:11:55 GMT], [Server, uvicorn]}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 369
```

---

## 🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl -X POST http://192.168.20.10:9000/process-file
```

Flow:

Upload → Processing → AI → Response returned across WAN

**ผลลัพธ์ :**

```
PS E:\Levto\Com Sci KKu\ปี 2 เทอม 2\Computer Netwrok\automation\week03-microservices\phase1> python -c "import requests; r = requests.post('http://localhost:9000/process-file', files={'file': ('sample.jpg', open('test_data/sample.jpg', 'rb'), 'image/jpeg')}); print(r.status_code); print(r.text)"
500
{"detail":"Workflow failed: 'status'"}
```

---

## 🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=11ms TTL=255
Reply from 192.168.10.1: bytes=32 time=2ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

```
C:\>ping 100.10.10.2

Pinging 100.10.10.2 with 32 bytes of data:

Reply from 100.10.10.2: bytes=32 time=1ms TTL=254
Reply from 100.10.10.2: bytes=32 time=1ms TTL=254
Reply from 100.10.10.2: bytes=32 time=1ms TTL=254
Reply from 100.10.10.2: bytes=32 time=2ms TTL=254

Ping statistics for 100.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=127ms TTL=126
Reply from 192.168.20.10: bytes=32 time=101ms TTL=126
Reply from 192.168.20.10: bytes=32 time=204ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 101ms, Maximum = 204ms, Average = 144ms
```

Verify NAT overload working on R1:

```
None
show ip nat translations
```

**ผลลัพธ์ :**

```
R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:40    192.168.10.10:40   203.0.113.2:40     203.0.113.2:40
icmp 203.0.113.1:41    192.168.10.10:41   203.0.113.2:41     203.0.113.2:41
icmp 203.0.113.1:42    192.168.10.10:42   203.0.113.2:42     203.0.113.2:42
icmp 203.0.113.1:43    192.168.10.10:43   203.0.113.2:43     203.0.113.2:43
```

---

# 🔎 Verification Commands

## On R1

```
None
show ip route
show ip nat translations
show ip interface brief
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/0/0
L       100.10.10.1/32 is directly connected, Serial0/0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 100.10.10.2
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     203.0.113.1     YES manual up                    up
GigabitEthernet0/1     192.168.10.1    YES manual up                    up
Serial0/0/0            100.10.10.1     YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

## On R2

```
None
show ip route
ping 100.10.10.1
```

**ผลลัพธ์ :**

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 100.10.10.1 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/0/0
L       100.10.10.2/32 is directly connected, Serial0/0/0
S    192.168.10.0/24 [1/0] via 100.10.10.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 100.10.10.1
```

```
R2#ping 100.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/16 ms
```

---

# Engineering Reflection (Week03 Integration)

From your Week03 profile :

This lab now demonstrates:

| Concept | Where Demonstrated |
|---|---|
| Public vs Private Boundaries | R1 NAT edge |
| Distributed Services | ServerA & ServerB |
| Network Segmentation | Serial ISP /30 |
| Horizontal Architecture | Two independent microservice stacks |
| Internet Edge Pattern | R1 as NAT Gateway |
| Service Isolation | LAN A vs LAN B |

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# What Changed from Lab 4

| Lab 4 | Lab 5 |
|---|---|
| 10.10.12.0/30 simulated internet | Real ISP serial 100.10.10.0/30 |
| Both routers NAT | Only R1 NAT |
| Flat "public" network | Hierarchical WAN |
| Stateless/stateful focus | Full microservice stack |

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# Architectural Significance

You now have:

- Enterprise Edge Router

- ISP Serial WAN
- Branch Router
- Dual Microservice Deployments
- NAT Boundary
- Cross-site distributed system

This is no longer a classroom NAT lab.

This is a **miniature cloud-edge + branch distributed architecture.**

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